

HOW ARE NEW MOLECULAR CRYSTALS **DESIGNED**?

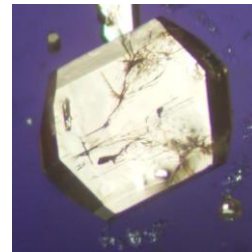
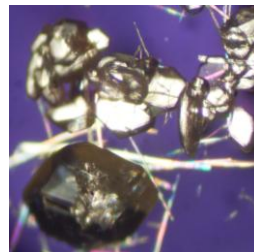
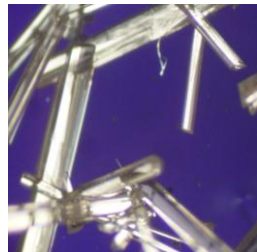
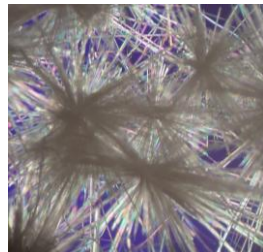
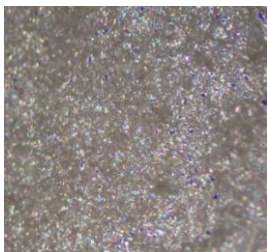
Design strategies

- high throughput crystallization
- rational design
- crystal structure prediction

HOW ARE THEY **SYNTHESISED**?

Supramolecular synthesis

- from solution (evaporations/cooling/diffusion/anti-solvent)
- mechano-chemical grinding (neat, liquid-assisted, cryogen)
- growth from the melt
- slurry



Remember CE's aim = design and control the packing of the molecules

High throughput crystallisation

- large & quick **screening**
- Industrial approach
- automation-derived

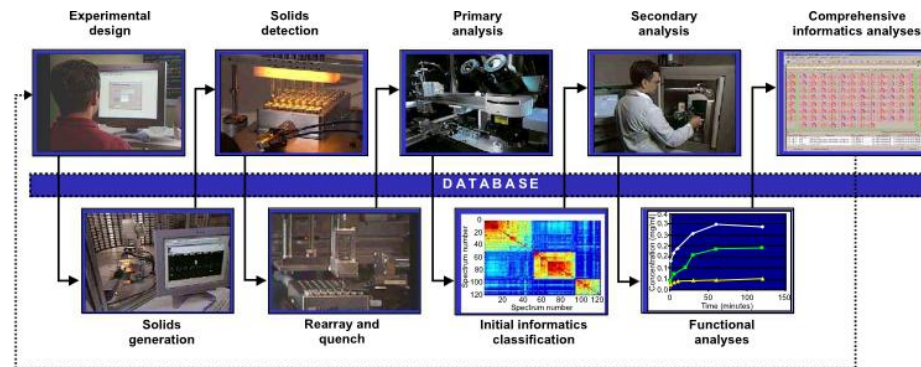
Used:

- Drug discovery
- Material science
- Agriculture
- Food safety
- Environmental analysis
- Chemistry to accelerate discovery and data collection

HT-screening

- the desired result is a better property or potency
- quick conduction of thousands to millions of tests
 - >>> chemical, genetic, pharmacological ..
- typical **hit rate 0.1%** of the total number of samples
- handful of compounds **from a selection of thousands**

workflow of a fully integrated HT crystallisation system

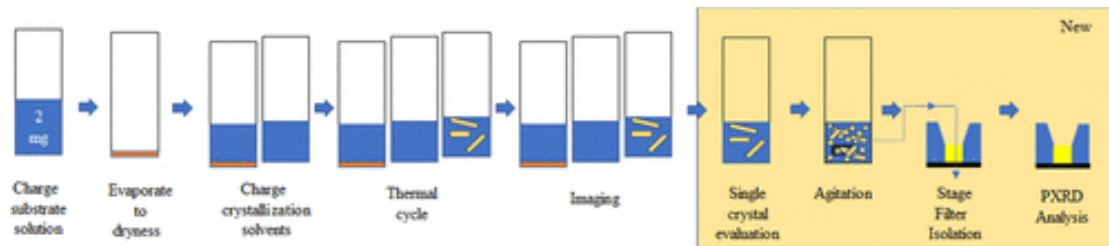


HT crystallisation

- desired formation of a solid.
- yield hit rates range from 10 - 100%, depends on the type of experiment and the process mode(s)
- small-scale (0.5–2 mg) crystalliser screening array
 - typically **10 – 50% trials may yield solids**
 - higher outcome

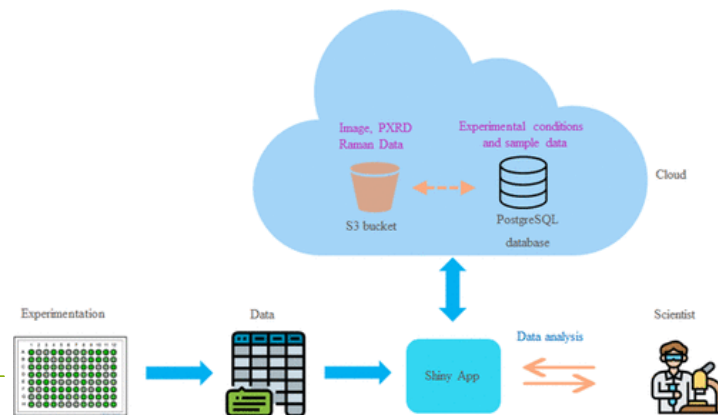
High throughput crystallisation

- smaller screening
- academic approach
- partial or no automatisisation
- critical selectivity of trials
- higher hit/sample number ratio



	Solvent	2:1 w/w water	1:2 w/w water	1:1 w/w MTBE	1:1 w/w Heptane*		Solvent	1:1 w/w MTBE	2:1 w/w Heptane	1:2 w/w Heptane		Solvent	2:1 w/w Heptane	1:2 w/w Heptane
acetonitrile						1 butanol					iPrOAc			
methanol						2 butanol					n-BuOAc			
ethanol						n-Amyl-OH					MTBK			
isopropanol						MEK					MTBE			
acetone						MeTHF					DCE			
THF						DME					Toluene			
1 propanol						1,4 Dioxane					Anisole			
None						EtOAc					CPME			

* Toluene is substituted for Heptane with Acetonitrile and Methanol



relies on

- computational modeling
- fundamental thermodynamics to predict and guide molecular packing, crystal habits, and physical properties

replaces traditional trial-and-error approaches

What does it mean?

- break down further into just a few systems
- understand similar cases
- understand intermolecular interactions
- translate the knowledge to choose best candidates
 - few targets
 - optimisation of conditions
 - structural analysis
 - structure-property correlation

Design Workflow

- Database Mining & Computational Screening
- Intermolecular Interaction Engineering

analysis of existing crystal structures via the CSD

- statistical analysis of packing motifs
- information on functional groups
- intermolecular interactions
- controlling specific non-covalent forces

- (Machine learning integration)

Liu et al. Cryst. Growth Des. 2026, 26, 2081–2094

Identification of supramolecular synthons or motif

crystal = supermolecule & intermol. interactions = glue

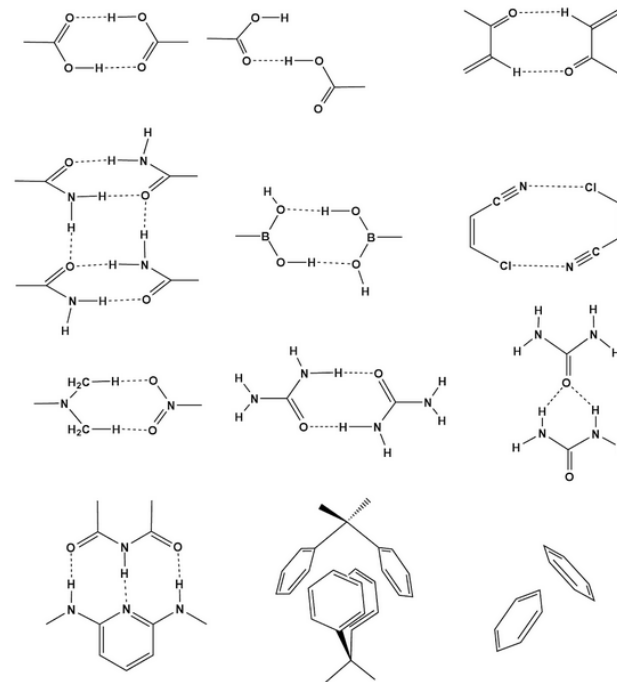
How to use “glue”?

- find a recurring packing pattern
- find a reliable functional group which makes a robust motive
- use it to create a new solid-state material



Supramolecular synthon of motif

- small repetitive units
- fully describe the essence of the structure
- define specific intermolecular interactions

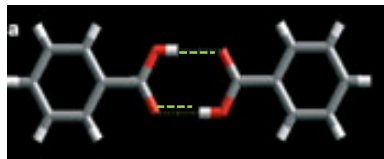
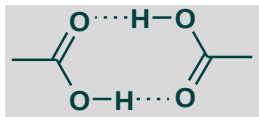


Some commonly used synthons.

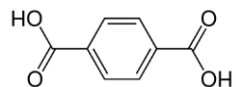
Simplification of complex systems describing crystals as **networks**

Nodes-Connectors Approach & Building up dimensionality

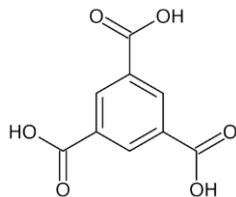
carboxyl dimer motif:



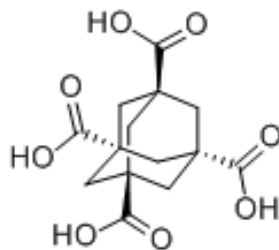
Nodes
and
Connectors



terephthalic acid



trimesic acid

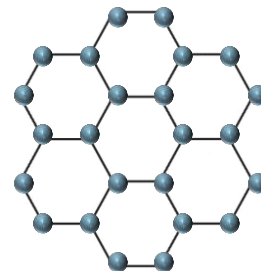


adamantane-1,3,5,7- tetracarboxylic acid

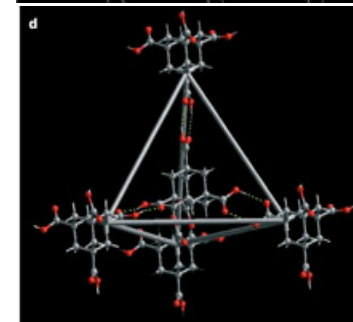
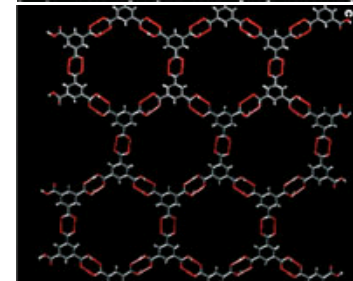
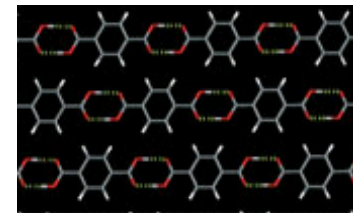
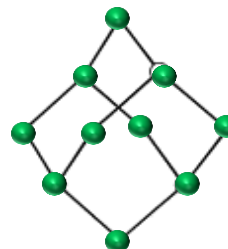
1D



2D



3D



1. Useful and reliable supramolecular motif

combines an **economy in size** with a **maximum of structural information**

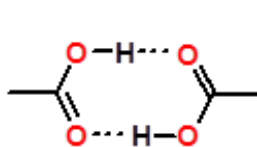
- small SS: very frequently - but no much specific information (O-H $\cdots$ O-H)
- large SS: plenty of structural information - but observed rarely, hardly accessible

2. Commonly used:

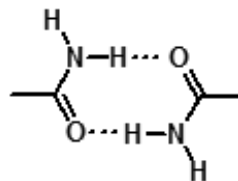
few functional groups held together by **strong and directional non-covalent interactions**

- Crystallisation - based on mutual recognition of molecules
- Molecular self-assembly with pre-desired arrangement

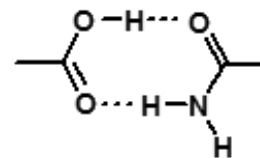
Different molecular systems can give very similar crystal structures and in opposite very similar molecules can crystallize in extremely different ways!



acid-acid
homodimeric synthon: **homodimer**



amide-amide



acid-amide
heterodimer

Combine an acid and an amide to form a co-crystal

What will happen?

- ⇒ the co-crystal contains homodimer
- ⇒ the co-crystal contains heterodimer

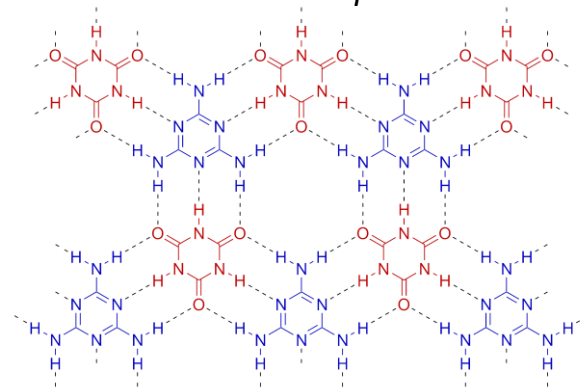
CSD Statistics: Heterosynthons occur more frequently than homosynthons

- “a key to predictability” (M. Zaworotko)
- successful co-crystallization relies on pairing molecules with matching, predictable intermolecular interactions

Three-centered heterosynthon
- **multi-point recognition pattern** –
in barbituric acid /melamine co-crystal

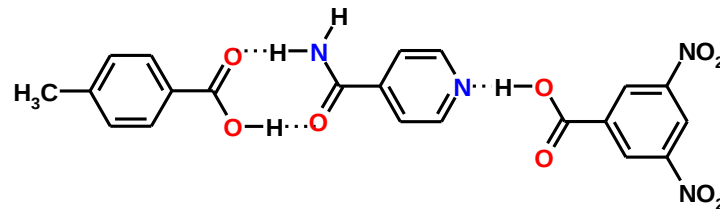


chicken-wire pattern



Expand to three components = Ternary co-crystal

Hierarchy of HB: Stronger acid interacts with the best acceptor (pyridine N) and the weaker acid with the amide.



4-methylbenzoic acid – isonicotinamide – 3,5-dinitrobenzoic acid

Computational Screening

Calculation of intermolecular potentials in crystal structures using:

UNI Force Field Calculations of Mercury crystallography package

Unified Pair Potential Model - semi-empirical set of intermolecular potentials

Scope and Limitations

Parameterization:

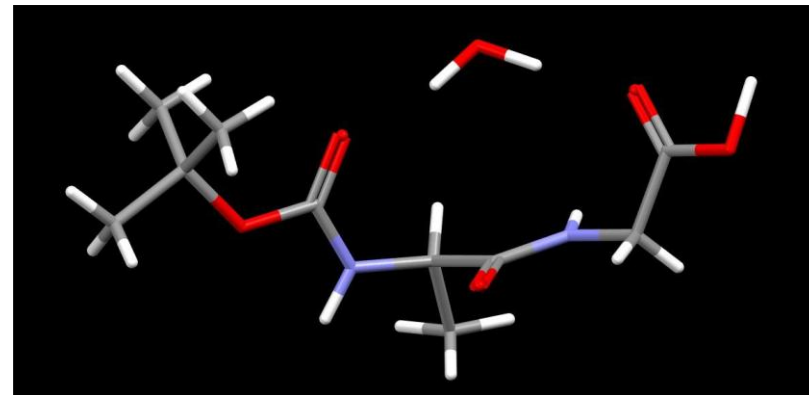
for a wide variety of elements common in organic chemistry:

Calculation Time & Costs:

- optimizations and calculations usually fast
- scaling roughly linearly with the number of atoms
- computationally inexpensive

Accuracy of intermolecular energy calculations using UNI

- semi-empirical and approximate
- lack the precision
- highly valuable for quickly assessing the role of different **hydrogen-bonding and aromatic stacking interactions**



SUKLEP
t-Butoxycarbonyl-L-alanyl-glycine monohydrate

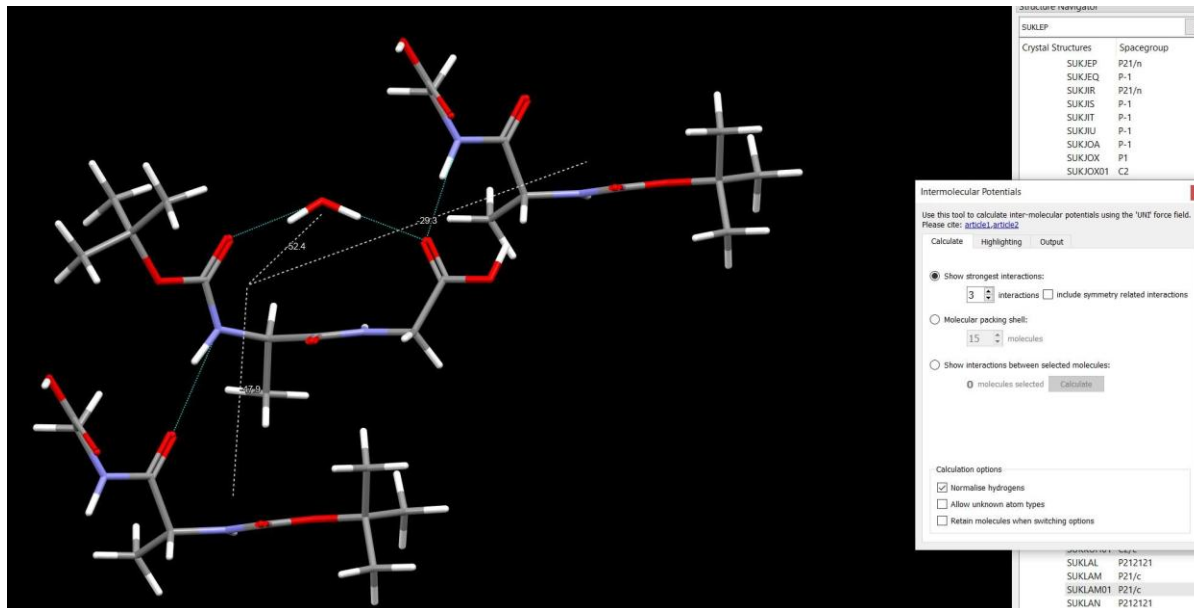
Computational Screening

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Unified Pair Potential Model - semi-empirical set of intermolecular potentials

- $\text{C}=\text{O} \cdots \text{H}_2\text{O}$: -52.4 kJ/mol
- $\text{NH} \cdots \text{O}=\text{C}$: -47.9 kJ/mol
- $\text{COOH} \cdots \text{HN}$: -29.3 kJ/mol



CSD-Database analysis with CCDC software package

UNI Force Field Calculations of Mercury crystallography package

Co-Crystal Prediction of Mercury crystallography package

Crystal Explorer for Hirshfeld Analysis and DFT calculation of intermolecular interactions energies

Useful supportive tools but not yet truly reliable

- need a combination of several methods for good results
- Often need extended understanding and parameter optimisation
 - BUT allow quick insights & contribute to rational design

Structure Interpretation:

Energetic vs. geometrical approach

Crystal is a solid form of a substance with a definite and **periodic** internal structure.

I. What is the essential, **basic structural motive (BSM)** in the crystal packing?

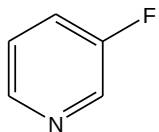
BSM – infinite structural fragment

II. Which types of **intermolecular interactions dominate** the molecular aggregation?

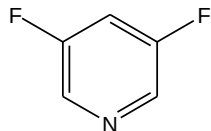
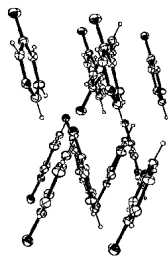
Possibilities for the interpretation of the crystal packing:

- **geometrical** -- visualization of the crystal packing
- **energetic** -- computational methods

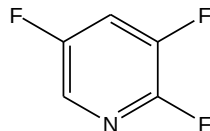
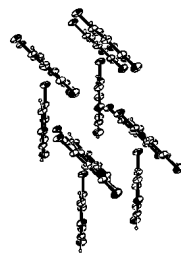
Crystal packing of fluoropyridines



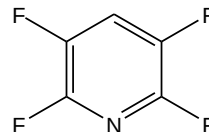
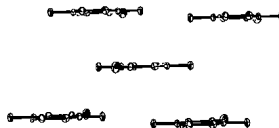
3-F-Py



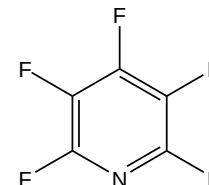
3,5-F-Py



2,3,5-F-Py



2,3,5,6-F-Py



penta-F-Py



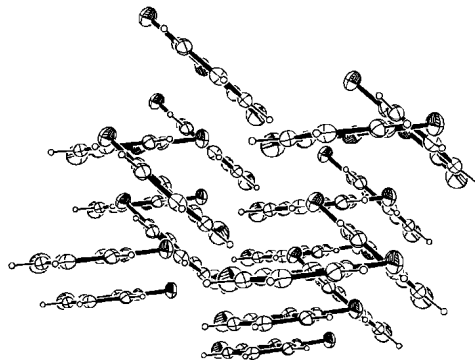
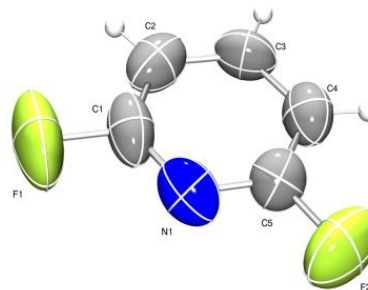
1. Planarisation with a growing number of fluorine substituents
2. Switching back to edge-to-face aggregation

Geometrical interpretation of crystal packing

➤ What You See Is What You Want

2,6-Difluoropyridine: geometrical interpretation

- distorted edge-to-face arrangement
- inter-ring angle: 46.8°
- centroid-centroid distance: $6.90 \text{ \AA}$
- **no** short F $\cdots$ F-, F $\cdots$ N-, F $\cdots$ H- or C $\cdots$ H-contacts



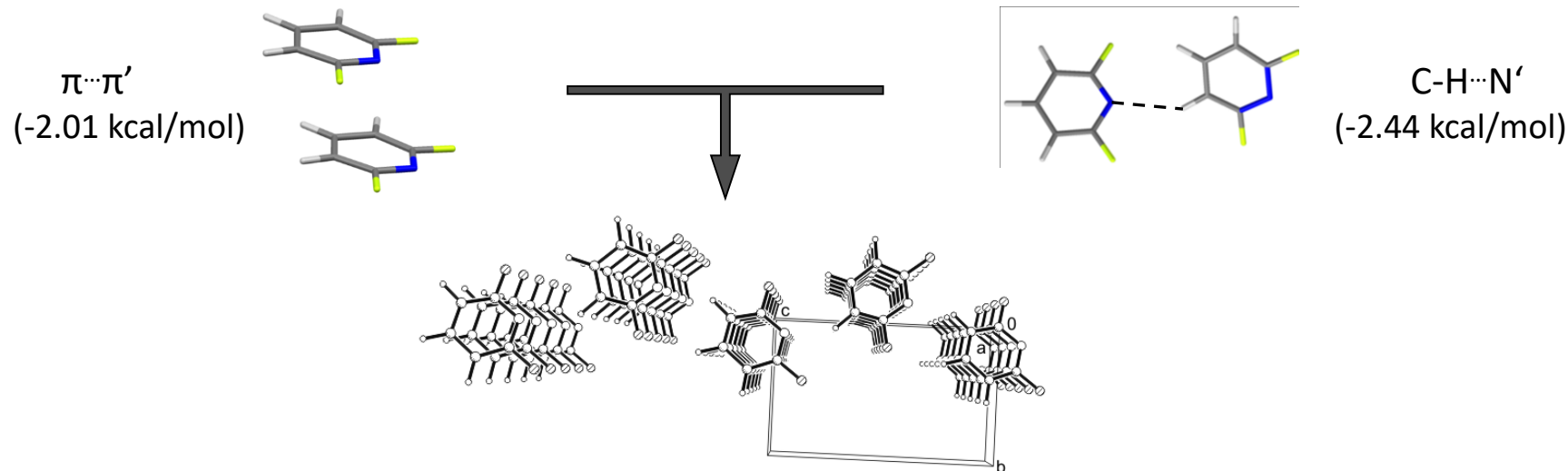
crystal packing overview

Energetical interpretation

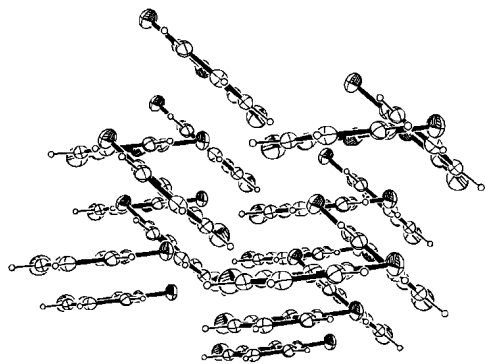
contribution of the energy of intermolecular interactions

- semi-empirical calculations with *ab-initio methods* (e.g. method/basis set: **MP2/ 6-311 G****)
- reliable calculation of high accurate
- based on experimental data (single crystal X-Ray analysis)

2,6-FP: Reconstructing crystal packing based on the most negative energy values

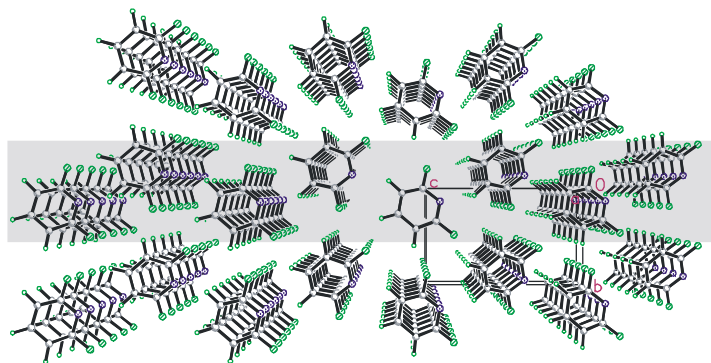


geometrical view



edge-to-face arrangement

energetic view

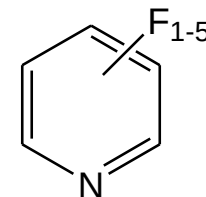


zig-zag layer

$$E_{\text{int}}^{\text{BSSEcorr}} = -10.67 \text{ kcal/mol}$$

$$E_{\text{ext}} = -2.11 \text{ kcal/mol}$$

Interactions: C-H $\cdots$ N & stacking



>> short F $\cdots$ F distances:
- smallest negative or
- even positive values

- from molecule to its structure from a scratch

NATURE VOL. 335 15 SEPTEMBER 1988

NEWS AND VIEWS

201

Crystals from first principles

A new calculation of the polymorphs of silica appears to have broken new ground in deriving crystal structure from chemical composition. But X-ray crystallographers need not worry — yet.

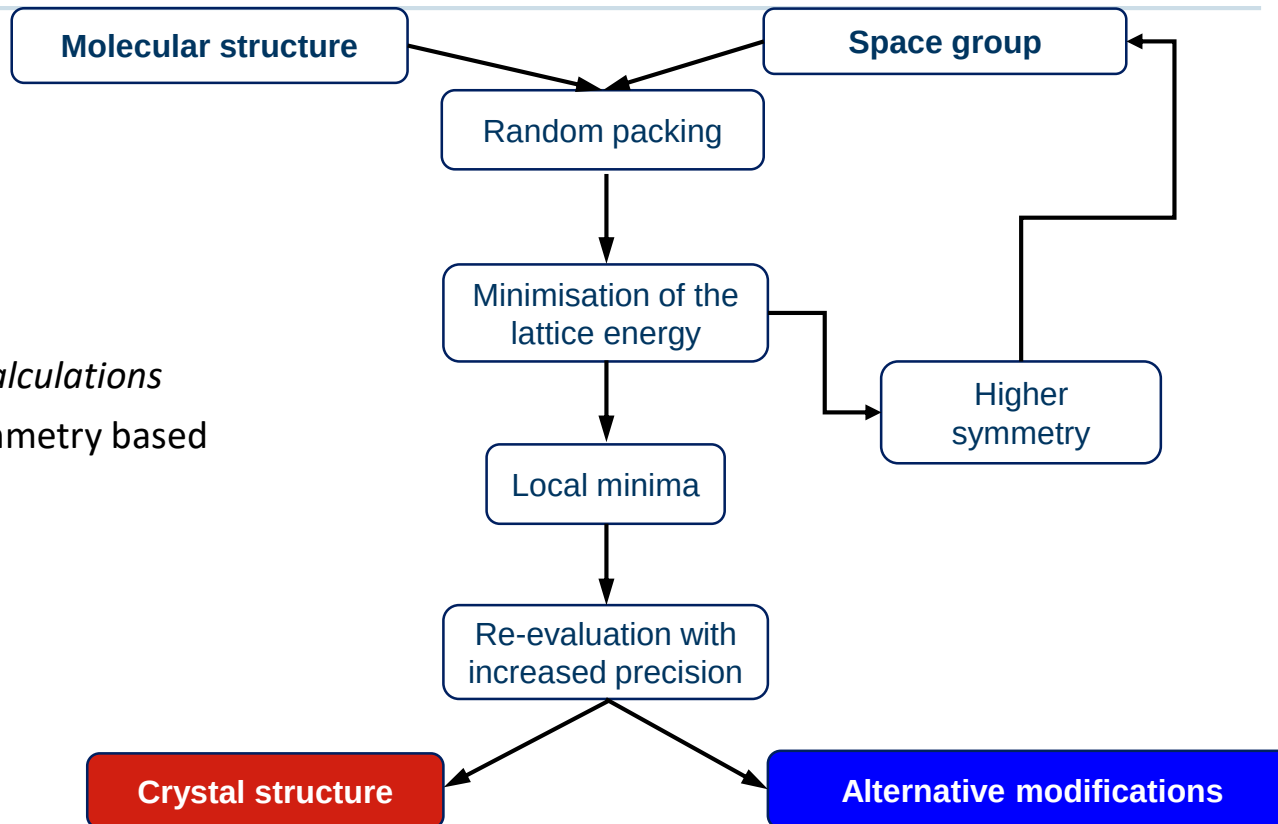
ONE of the continuing scandals in the physical sciences is that it remains in general impossible to predict the structure of even the simplest crystalline solids from a knowledge of their chemical composition. Who, for example, would guess that graphite, not diamond, is the thermodynamically stable allotrope of carbon at ordinary temperature and pressure? Solids such as crystalline water (ice) are still thought to lie beyond mortals' ken.

oxygen atoms — the authors estimate 0.7 of an electron charge for every SiO bond, or close on two electron charges for those O atoms bound to two Si atoms. So, to ensure that their calculated structures are electrically neutral, the authors place electronic charges at points 1.65 Å along the four tetrahedral axes of the cluster, neatly simulating in the process the electrostatic effects of the neighbouring oxygen ions found in structures such as

modulus within 10 per cent. The agreement between calculated energy (enthalpy) of the polymorphs and the few measurements available is even better (1 per cent), but the authors say they could probably do even better if they allowed for the band structure of these polymorphs and the extent to which the bands are filled at different temperatures.

Several refinements remain to be explored. The chances are that three-body potentials and even more complicated

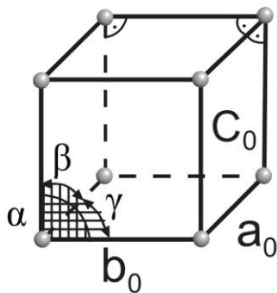
Crystal Structure Prediction approach: theoretical calculation



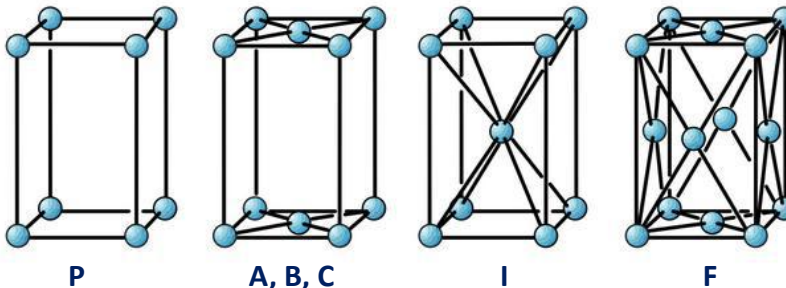
- *ab-initio* calculations
- crystal symmetry based

7 crystal systems

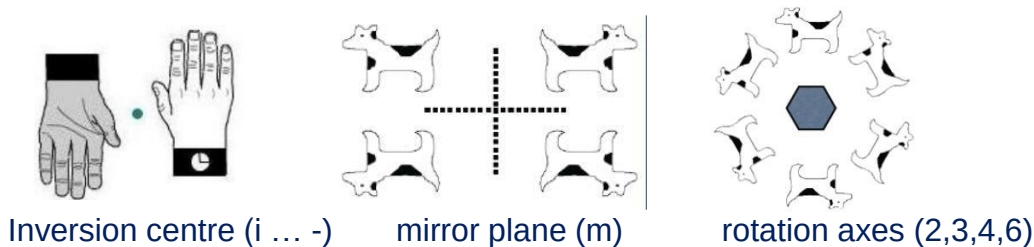
- Triclinic a, b, c variabel; α, β, γ variabel
- Monoclinic a, b, c variabel; $\alpha = \gamma = 90^\circ$; β variabel
- orthorhombic a, b, c variabel; $\alpha = \beta = \gamma = 90^\circ$
- Tetragonal $a = b, c$ variabel; $\alpha = \beta = \gamma = 90^\circ$
- Trigonal, hexagonal $a = b, c$ variabel; $\alpha = \beta = 90^\circ$; $\gamma = 120^\circ$
- Cubic $a = b = c$; $\alpha = \beta = \gamma = 90^\circ$



14 Bravais-lattices

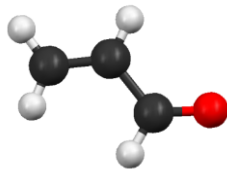
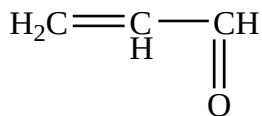


Symmetry elements with or without translation



Spacegroup = Combination of crystal systems, bravias lattice + symmetry + translation

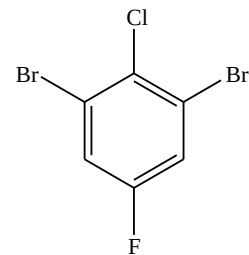
Crystal Structure Prediction (CSP): blind tests of CCDC



Acrolein

4th blind test, 2009

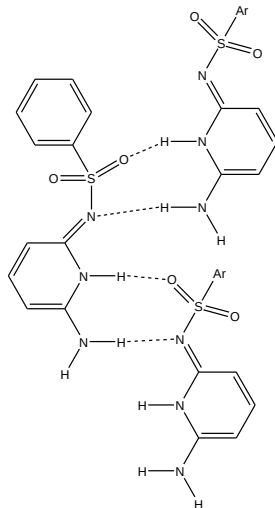
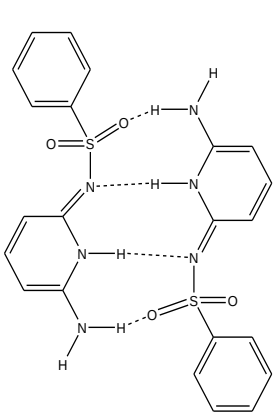
Success for small rigid molecules



1,3-dibromo-2-chloro-5-fluorobenzene

(a)

(b)



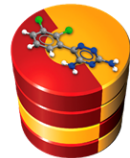
2nd blind test, 2002

6-amino-2-phenylsulfonylimino-11,2-dihydropyridine
four-point motif (a)
two-point motif (b)

No proper prediction, because of
Flexibility & different interaction modes

6. Blind Test

➤ start 2015



The Cambridge Crystallographic
Data Centre

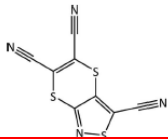
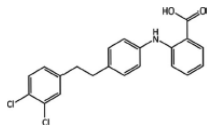
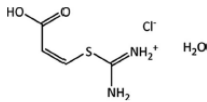
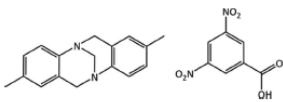
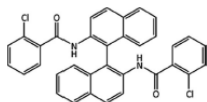
hhu
Heinrich Heine
Universität Düsseldorf

- (1) Rigid molecules, max. 30 atoms.
- (2) Partially flexible molecule with 2–4 internal degrees of freedom, max. 40 atoms.
- (3) Partially flexible molecule as a **salt**, 1-2 internal degrees of freedom, max. 40 atoms.
- (4) Multiple partially flexible (1-2degrees of freedom) independent molecules:
→ a co-crystal or solvate; max. 40 atoms.
- (5) Molecule with 4–8 internal degrees of freedom, 50–60 atoms.

→ *Each theoretical **research group** can suggest **3 best ranking predictions***

6. Blind Test

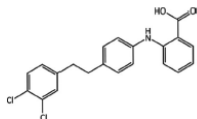
➤ published 2016, 21 groups, 90 researchers

Target	Chemical diagram	Crystallization conditions, remarks and clarifications	Attempted predictions	Times generated	Best rank (incl. re-ranking)
(XXII)		Crystallized from an acetone/water mixture; chiral-like character due to potential flexibility of the six-membered ring, but no chiral precursors used in synthesis.	21	12	1
(XXIII)		Five known polymorphs (A–E); three $Z' = 1$ (A, B, D), two $Z' = 2$ (C and E). The most stable polymorphs at 257 and 293 K are both $Z' = 1$. Crystallization conditions include slow evaporation of acetone solution and of ethyl acetate: water mixture.	A, B and D: 14; C and E: 3	A: 4, B: 8, C: 1, D: 3, E: 0	A: 23, B: 1, C: 6, D: 2, E: –
(XXIV)		Crystallized from 1 M HCl solution. The substituents of the C=C double bond are in the <i>cis</i> configuration.	8	1	2
(XXV)		Slow evaporation of a methanol solution, which contained a racemic mixture of the enantiomers of Tröger's base.	14	5	1
(XXVI)		Slow evaporation from 1:1 mixture of hexane and dichloromethane. No chiral precursors used in synthesis.	12	3	1

6. Blind Test 2015

- a difficult target

(XXIII)



Five known polymorphs (A–E); three $Z' = 1$ (A, B, D), two $Z' = 2$ (C and E). The most stable polymorphs at 257 and 293 K are both $Z' = 1$. Crystallization conditions include slow evaporation of acetone solution and of ethyl acetate: water mixture.

A, B and D: 14; C and E: 3

A: 4, B: 8, C: 1, D: 3, E: 0

A: 23, B: 1, C: 6, D: 2, E: –



Molecular conformations found in forms A–D of (XXIII), overlaid onto the fenemate group of the molecule; form A is in blue, form B in grey, form C molecule 1 is in red, form C molecule 2 in purple and form D in orange. H atoms are omitted for clarity.

- *slight differences between the polymorphs*
- *conformational changes*
- *flexibility of the phenyl group*

HOW ARE NEW MOLECULAR CRYSTALS **DESIGNED**?

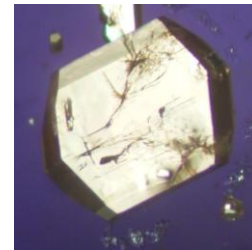
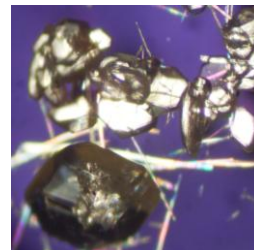
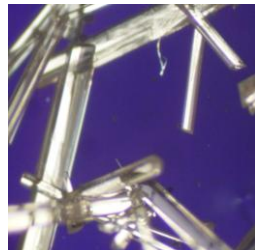
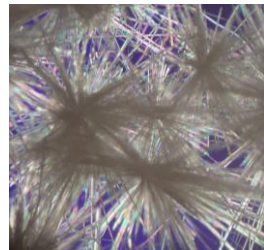
Design strategies

- high throughput crystallization
- rational design
- crystal structure prediction

HOW ARE NEW MOLECULAR CRYSTALS **SYNTHESISED**?

Supramolecular synthesis

- from solution (evaporations/cooling/diffusion/anti-solvent)
- mechano-chemical grinding (neat, liquid-assisted, cryogen)
- growth from the melt
- slurry



Remember CE's aim = design and control the packing of the molecules

One step – 2 results: **purification** and **particle formation**

- 1st order phase change
- solid product is isolated
- from solution, melt, vapour, solid phase

Supramolecular synthesis:

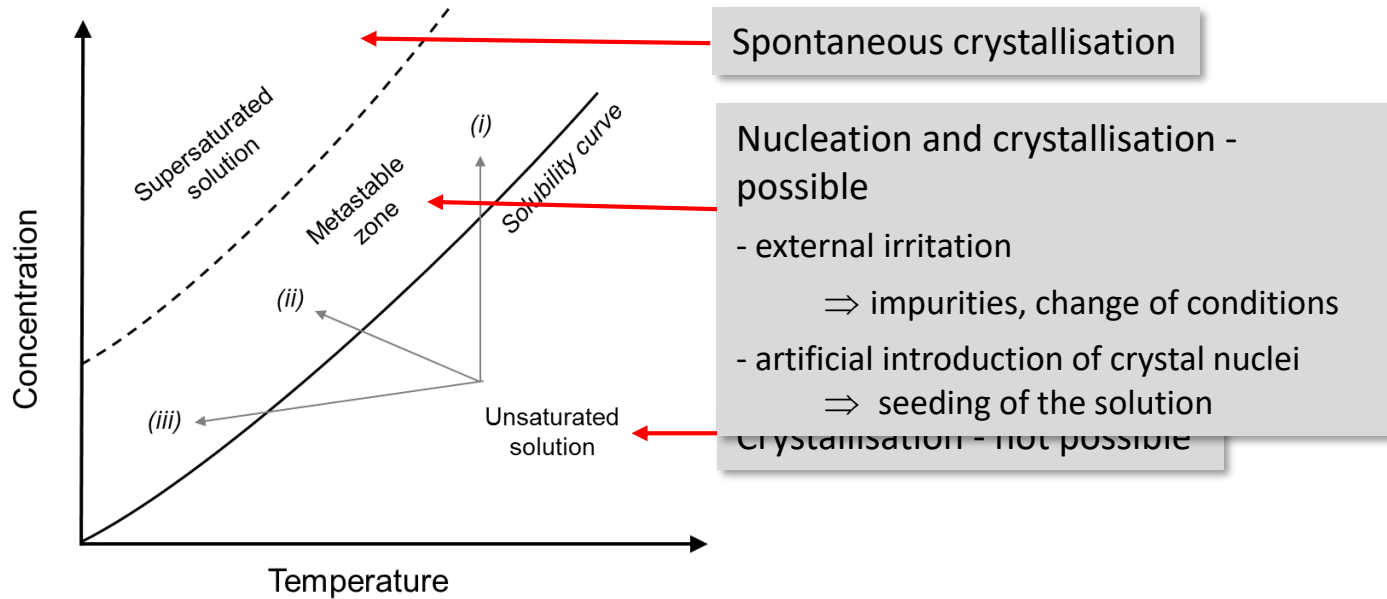
- Solution crystallization (evaporation)
- Antisolvent crystallization (inert solvent - decrease solubility)
- Melt crystallization (melt cooling, low temp. crystallisation)
- Hydrothermal and Solvothermal crystallisation (solvent, temp. > b.p.)
- Sublimation (vapour)
- Slurry (suspension)
- Mechanochemistry (grinding)

- ⇒ Supersaturation – for C. from a solution
- ⇒ Supercooling – for C. from a melt or liquid
- ⇒ High energy impact

Crystallisation from solution

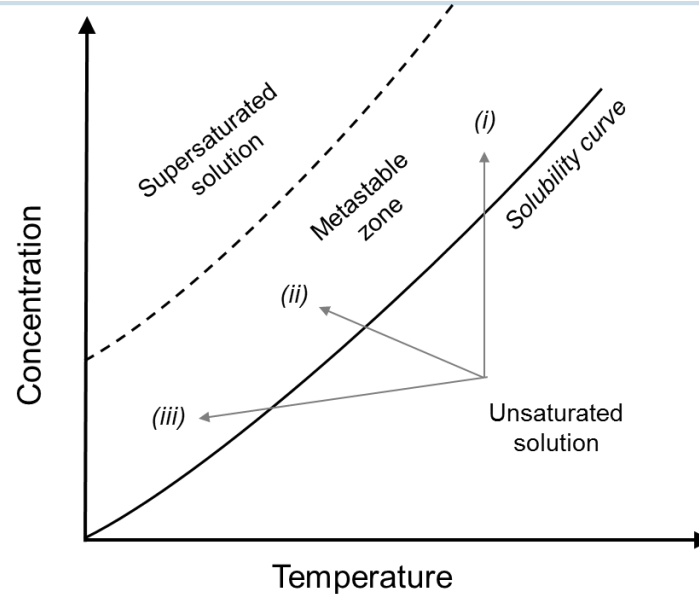
➤ Solubility diagram

variation temperature/concentration in relation to saturation and solubility



Crystallisation from solution

➤ Solubility diagram

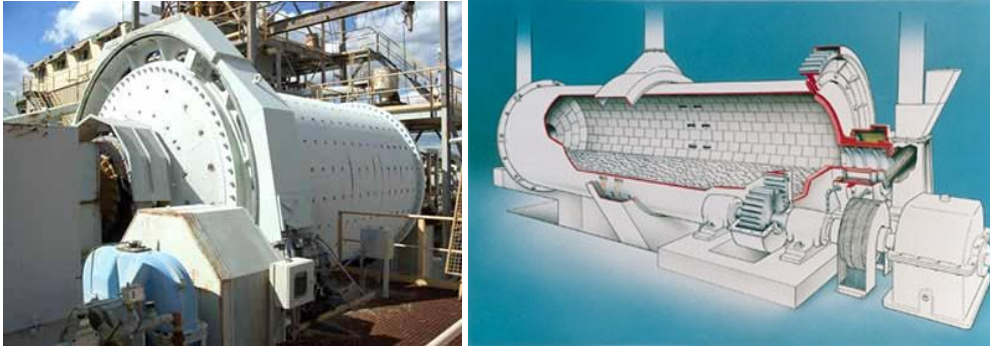


Crystallization from solution can be achieved by :

- (i) decreasing the solubility of the solid at constant temperature
- (ii) decreasing the vapor pressure of the solvent under reduced pressure
- (iii) cooling the solvent

Mechanochemistry

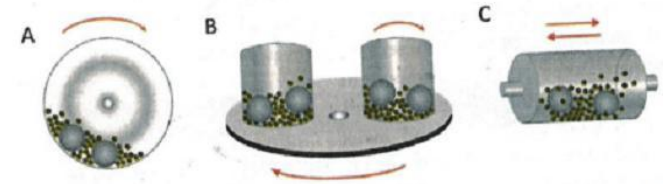
➤ Ball mills for grinding



Industrial application



Lab equipment



A: Kugelmühle, B: Planetenmühle, C: Schwingmühle

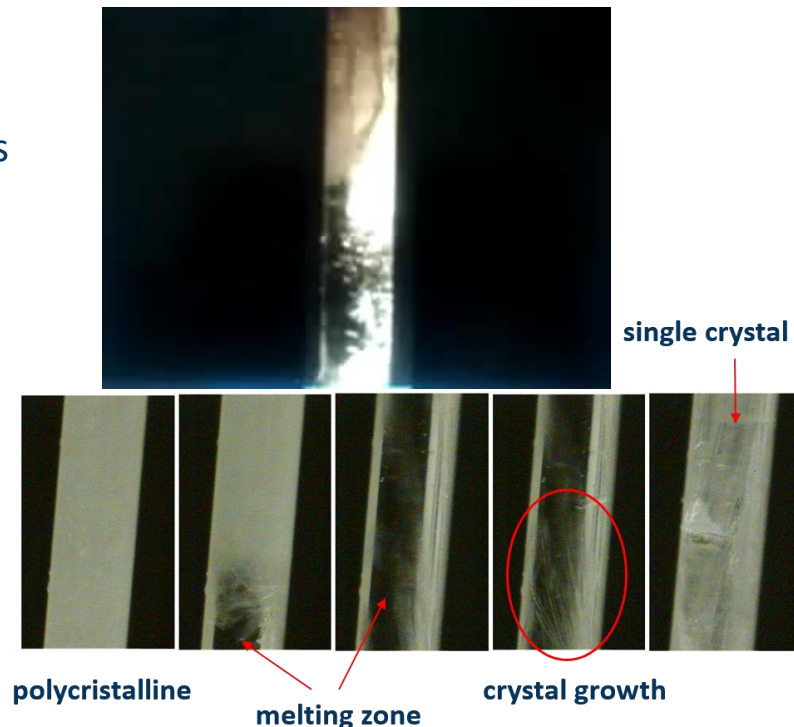
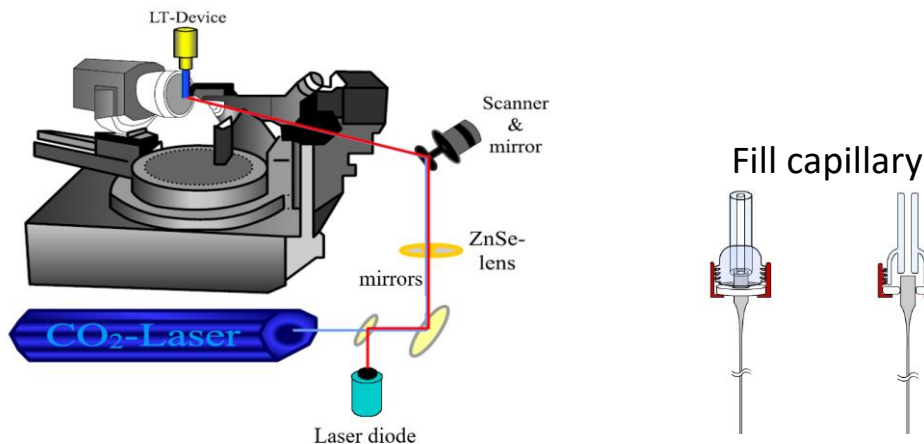
Grinding:

- Neat
- Liquid assisted (LAG)
- Cryogenic

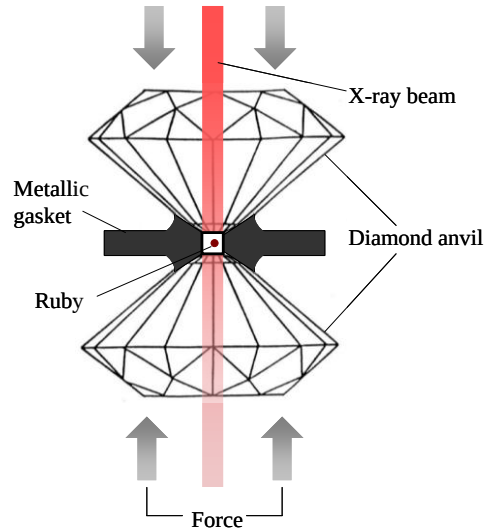
An exotic technique

in situ melt crystallisation at low temperature

- liquids
- low melting compounds like ionic liquids
- low temp. polymorphs
- co-crystallisation
- structural information of hardly accessible substances



Diamond Anvil Cell (DAC)



ruby: photometric determination
of the pressure inside the cell

Merril-Bassett-type DAC

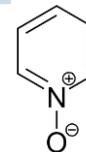


DAC prepared for X-Ray
measurment



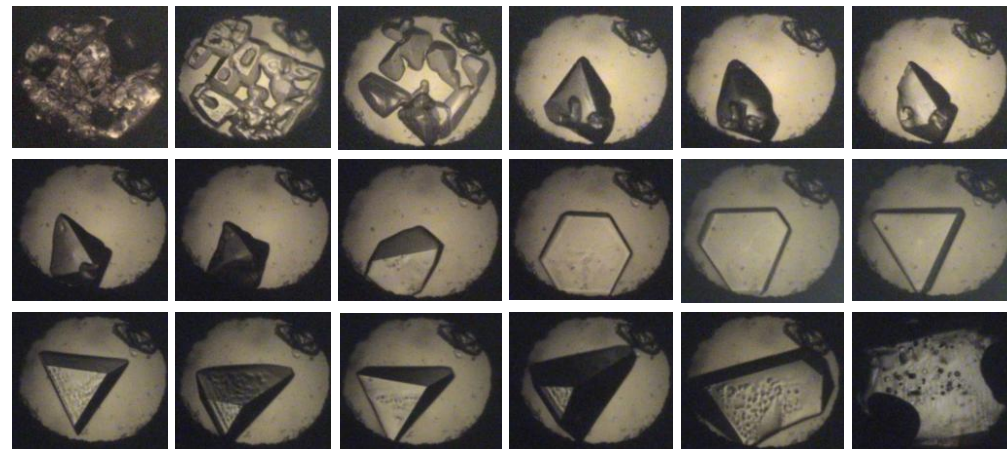
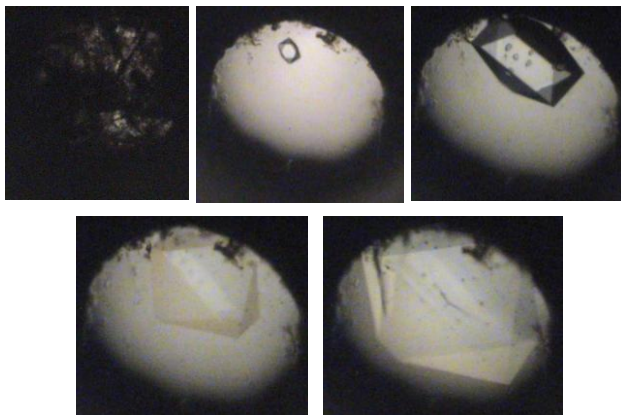
High pressure crystallisation of pyridine *N*-oxide

in a Diamond Anvil Cell



DAC with **beryllium disks** – 150°C limitation

$T_m = 65^\circ \text{C}$



$T_m = 90^\circ \text{C}$

Reorientation of the crystal

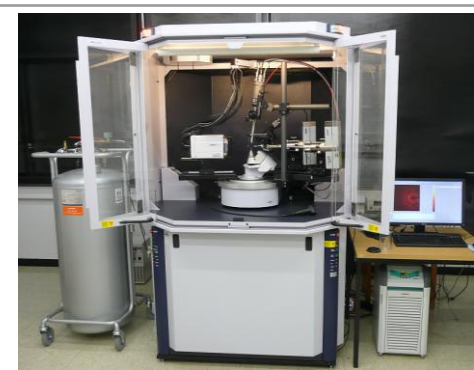
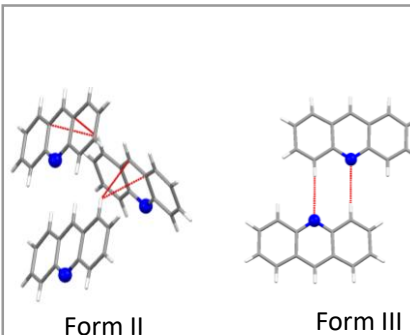
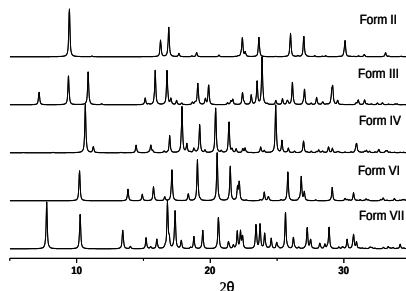
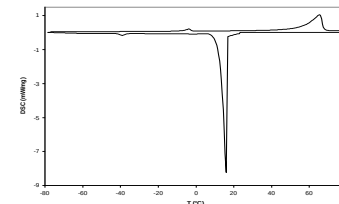
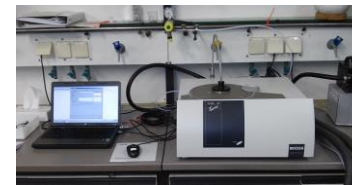
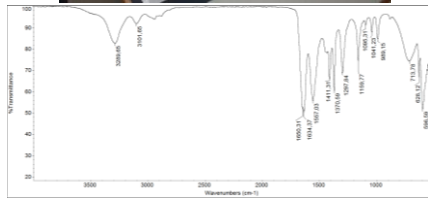
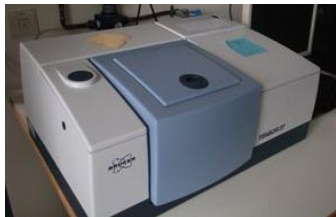
$P = 0.2 \text{ GPa}$

Solid-solid phase
transition ($T = 60^\circ \text{C}$)

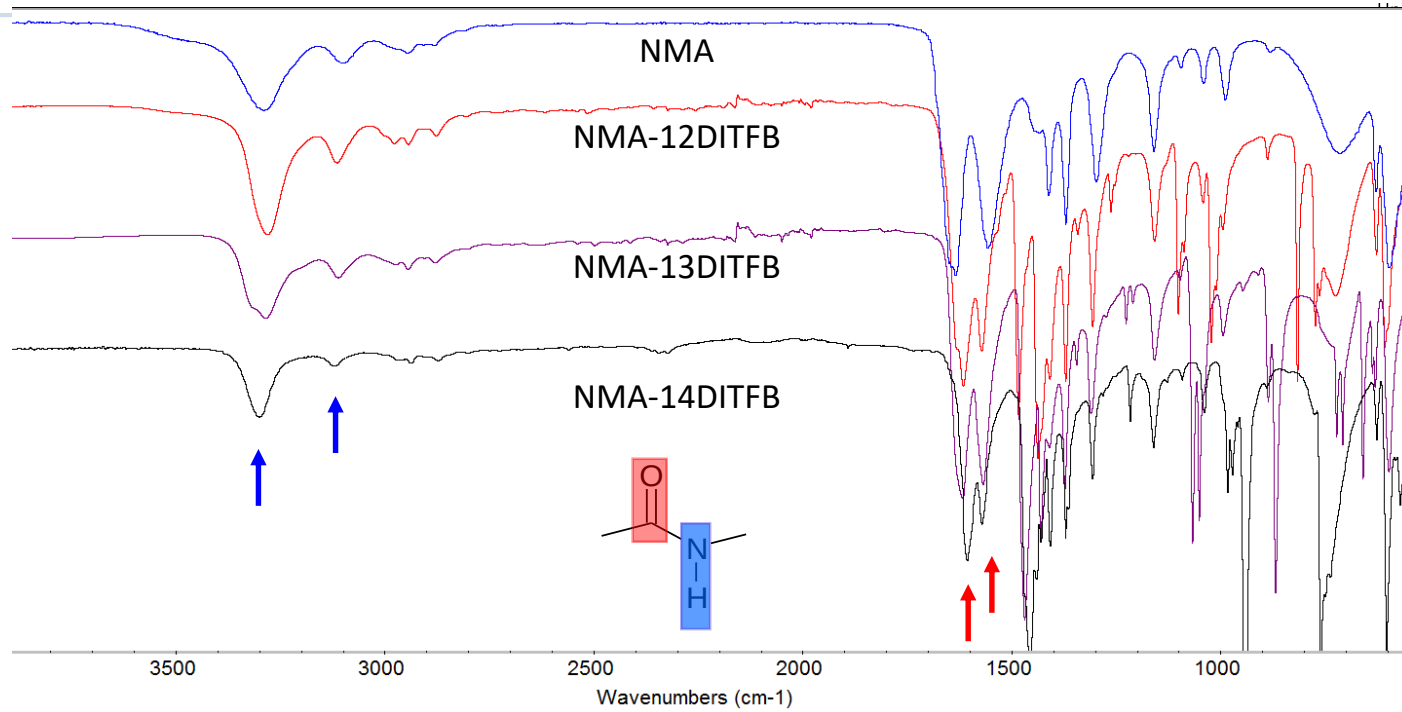
Methods of Characterization

➤ Crystals, Polymorphs, Co-crystals

- Spectroscopy – **IR** and **Raman**
- Thermal methods – **DSC** and **TGA**
- Powder X-Ray Diffraction – **PXRD**
- Single Crystal X-Ray Diffraction – **SC-XRD**



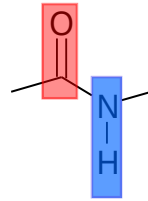
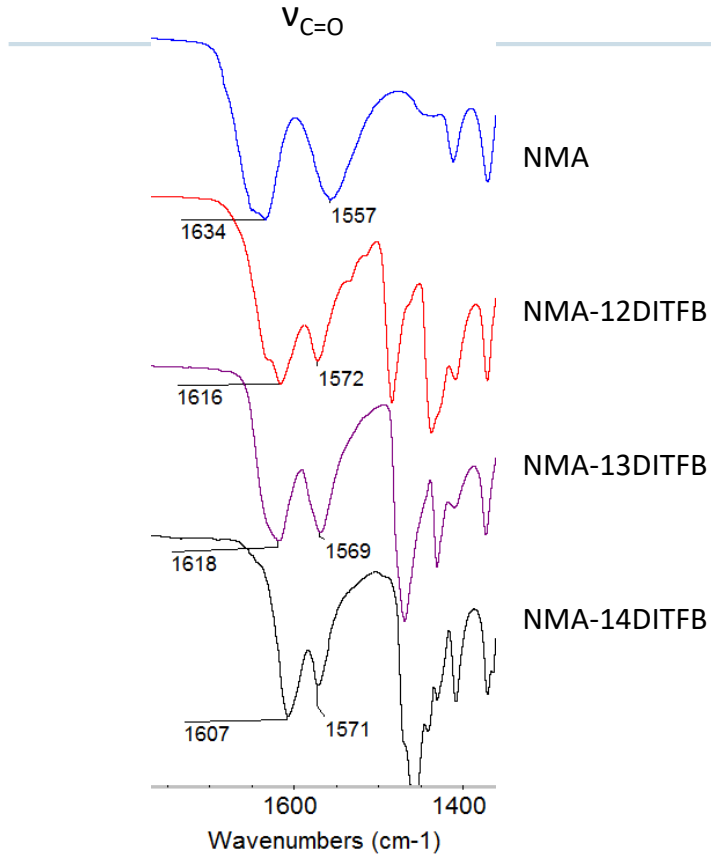
IR-spectra: NMA / DITFB



Shifts of the N-H and C=O vibration bands

IR shifts

NMA-DHaITFB-Systems



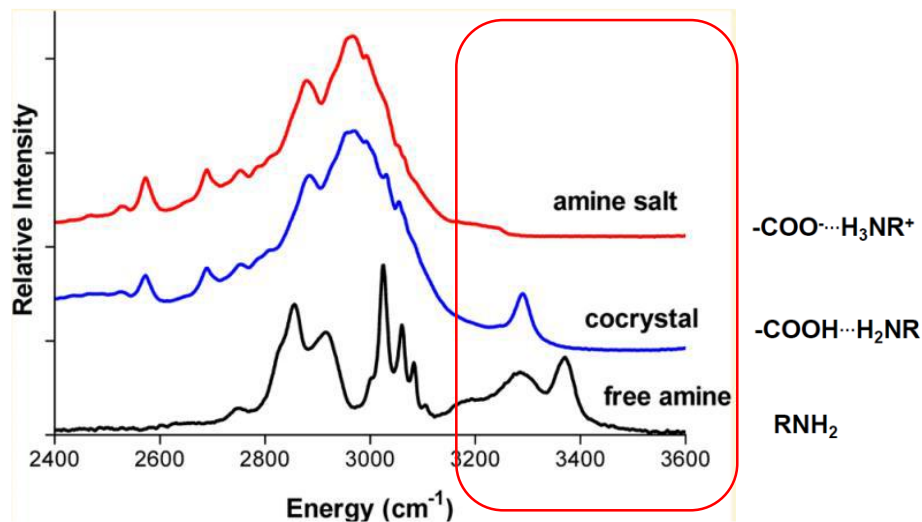
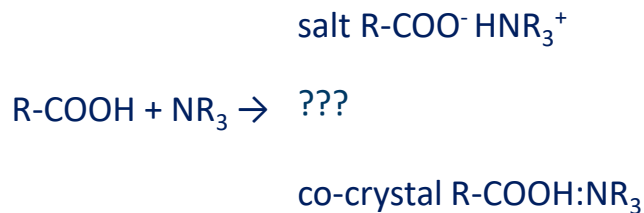
	$\nu_{\text{C=O}}$ (cm ⁻¹)	$\nu_{\text{N-H}}$ (cm ⁻¹)
NMA	1634	3102
NMA-12DITFB	1616	3115
NMA-13DITFB	1617	3112
NMA-14DITFB	1607	3118

- blue shift of the N-H band
- red shift of the C=O band



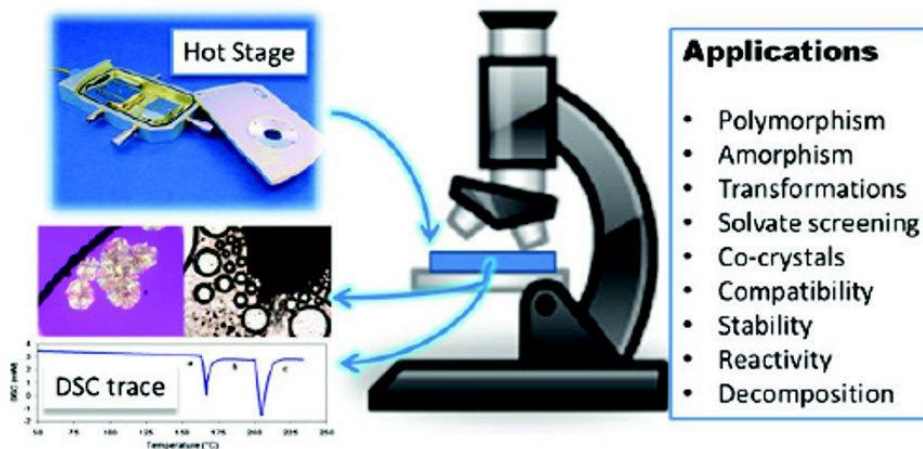
IR for MC-systems with ionic character

➤ distinguish between “salt” vs “co-crystal” formation



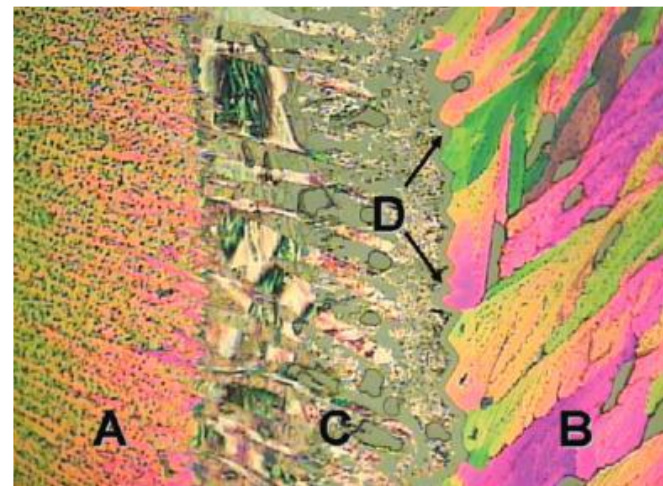
How can we prove polymorphs and MCCs?

Hot Stage Microscopy (HSM)



Contact area at 205°C

- A and B: two components
- C: **co-crystal**
- D: eutectic C:B starts to melt



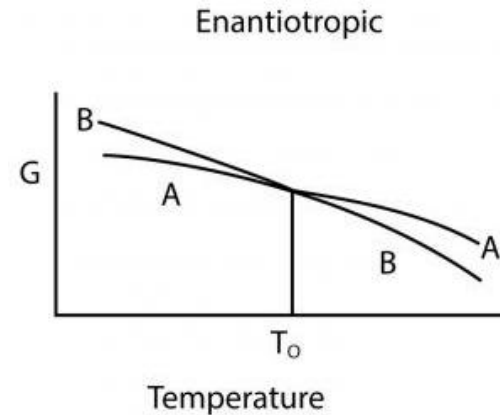
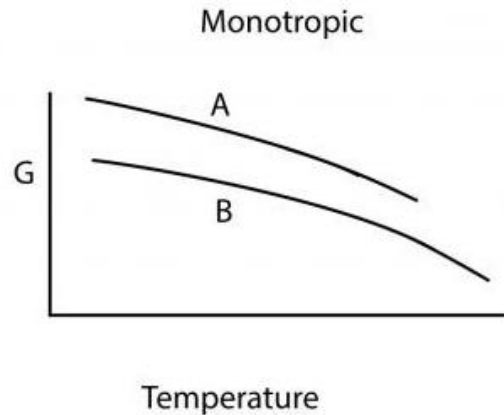
Monotropic vs. enantiotropic crystallisation

Monotropic polymorphic system

- Those for which only one form is stable, irrespective of temperature and pressure

Enantiotropic polymorphic system

- Different forms are stable under different external conditions

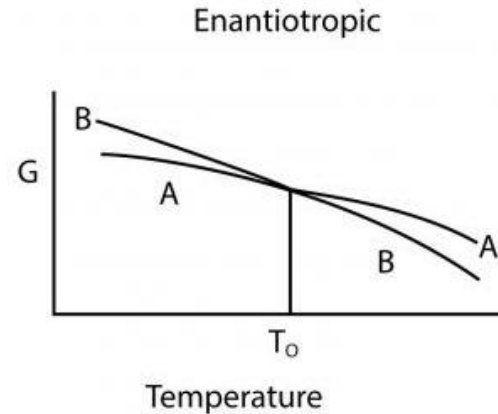
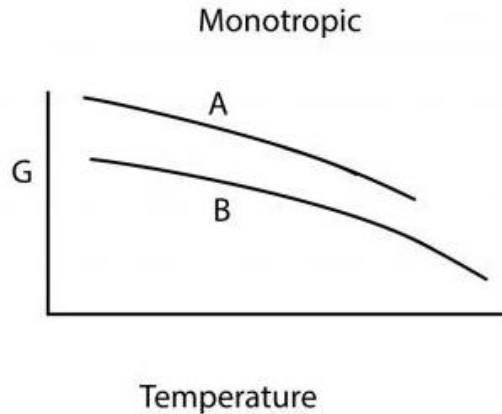


Polymorphs A and B

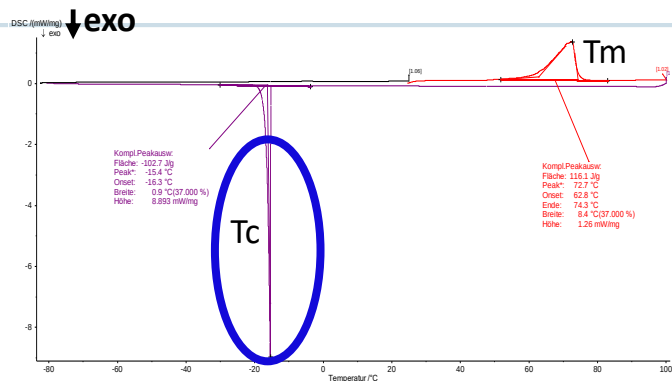
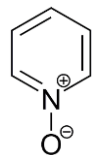
Monotropic system- polymorph B is more stable

Enantiotropic system – each enantiotrope has its own T-range of stability

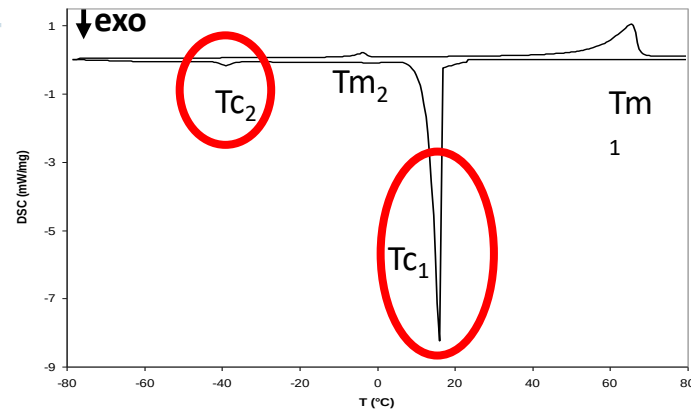
T_0 – transition temperature; both A and B are stable; equal free energy G



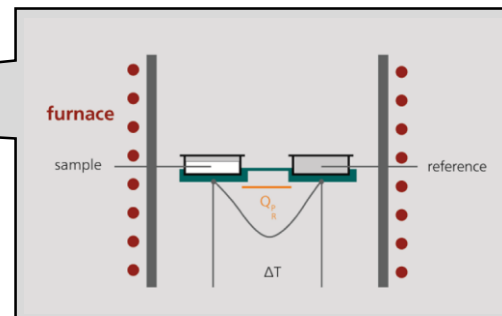
DSC: H/D-Pyridine *N*-oxide Polymorphism



Pyridine *N*-oxide
Form I



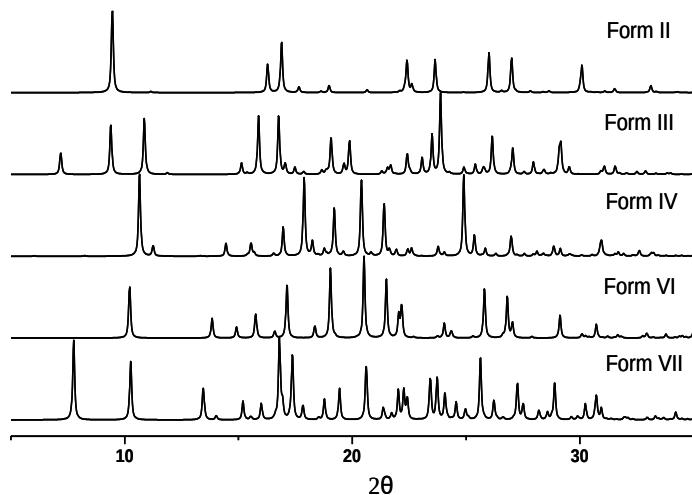
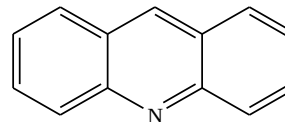
2,3,6-Trideuteropyridine *N*-oxide
Forms I & II



Powder X-Ray diffraction

➤ for Polymorphism check

acridine

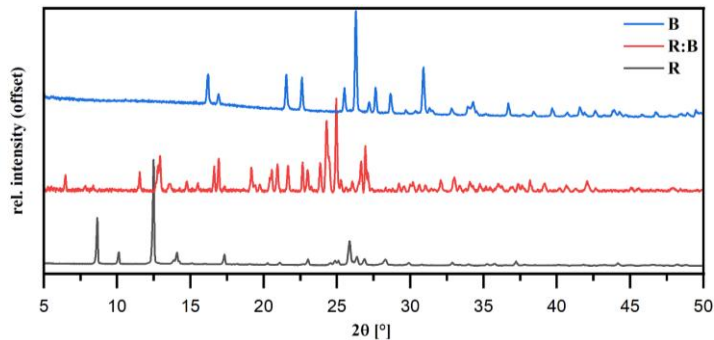


Different PXRD pattern for different forms

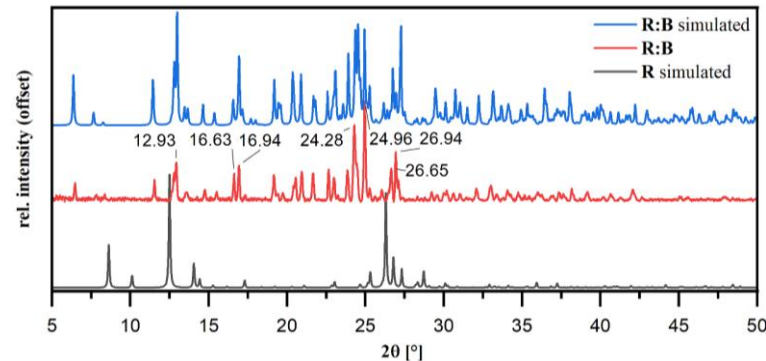
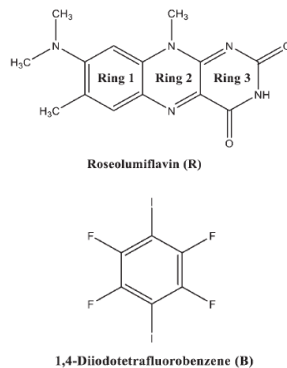
- Identification of new polymorphs
- Identification of mixtures

Powder X-Ray diffraction

➤ for co-crystal identification



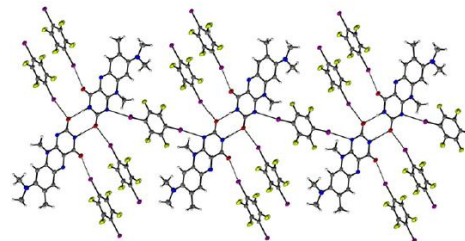
Experimental powder pattern of **R:B**
compared to experimental patterns of **R** and **B**



Experimental patterns of **R:B** compared to
simulated **R** and **R:B** from single crystal

Different PXRD pattern for new MCCs

- Identification of new co-crystals
- Phase purity and crystallinity via comparison with simulated powder pattern from single-crystal data





Si-SC-sample holder

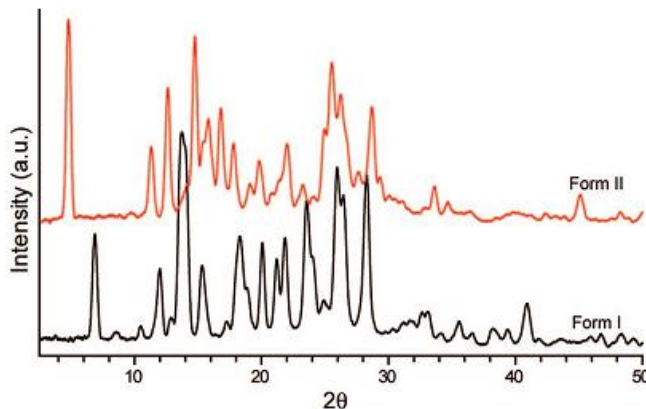


Figure 3. Powder X-ray diffraction patterns of CBZ-SAC I and CBZ-SAC II.

Differences in crystal **morphology**:

- plates (Form I)
- needles (Form II)

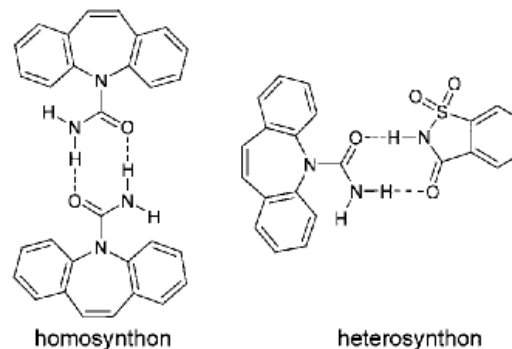
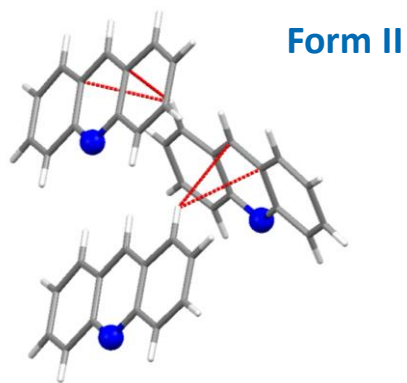
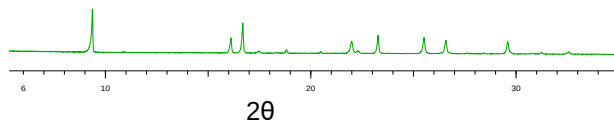


Figure 5. Representation of the homomeron between two CBZ molecules in CBZ-SAC I and the heteromeron between a CBZ and SAC molecule in CBZ-SAC II.

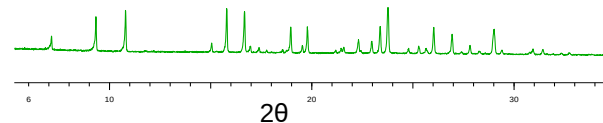
SC-XRD: H/D-Acridine

Acridine in acetone

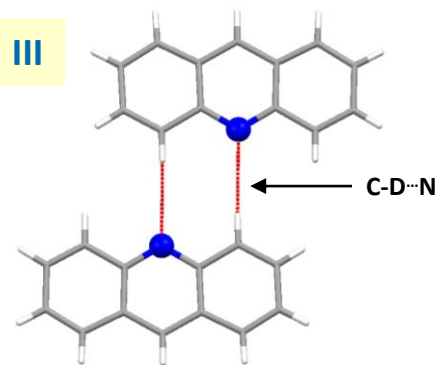


the most stable polymorph

d_9 -Acridine in acetone



Form III



stable at room temperature

⇒ **Stabilization of the less stable form III by N-D...C intermolecular interactions**